Add Custom Slots to a Lex Chatbot

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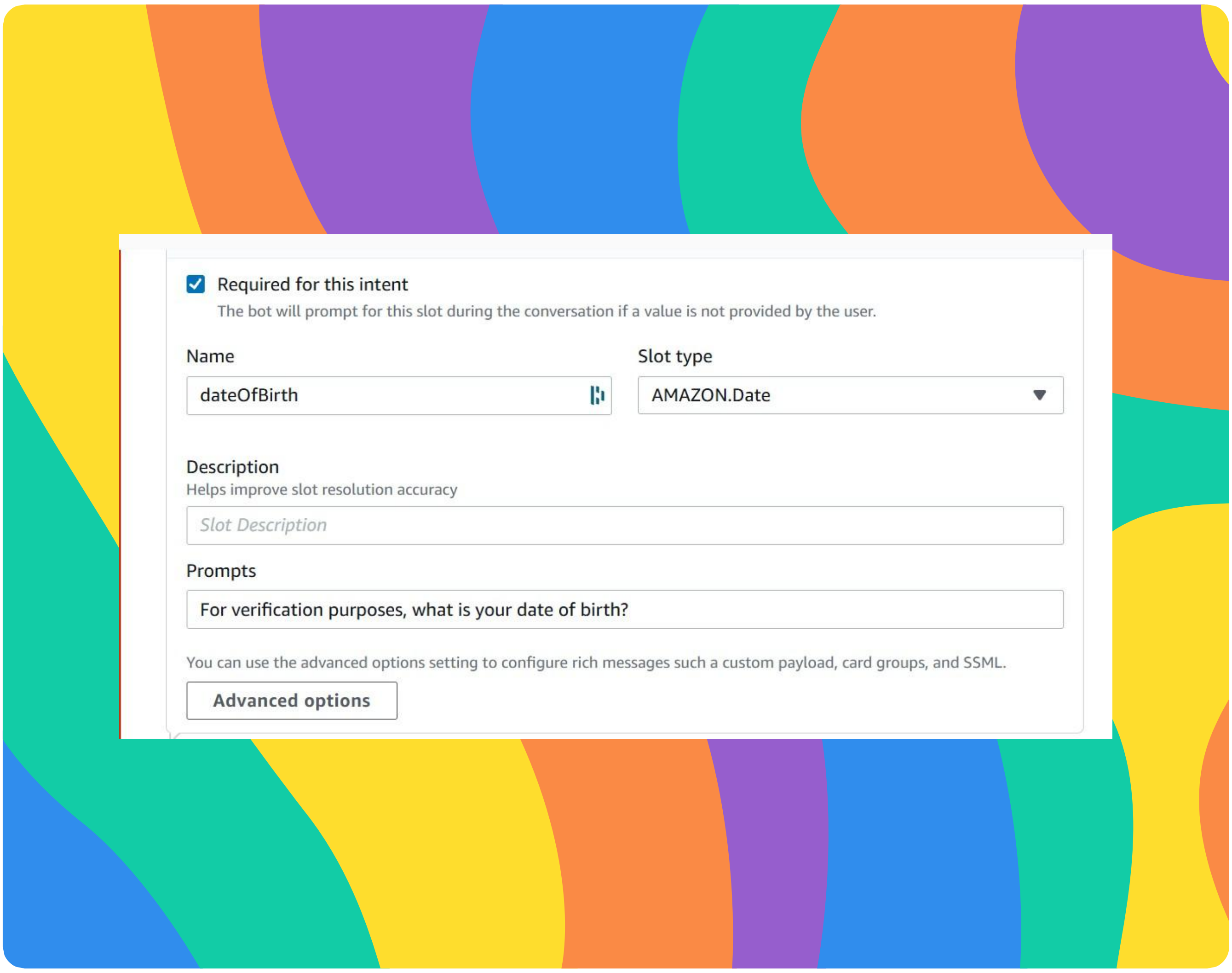


Introducing Today's Project!

In today’s project, I used Amazon Lex to build a conversational banking chatbot that can understand user requests and guide them through checking their account balance. I created intents, custom slot types, slot prompts, and failure responses so the bot could collect structured information like the user’s account type and date of birth. Lex handled the natural-language interpretation, slot filling, and dialog flow, allowing me to focus on designing a realistic, multi-turn conversation.

What is Amazon Lex?

For this project, Amazon Lex is the service I used to build a conversational banking chatbot. It provides the natural-language understanding and dialog management needed to interpret user requests, collect structured data through slots, validate inputs, and guide the user through multi-turn conversations like checking their account balance.

One thing I didn't expect in this project was...

I didn’t expect how smoothly Lex handled the conversation once I customized the slots and prompts.

This project took me...

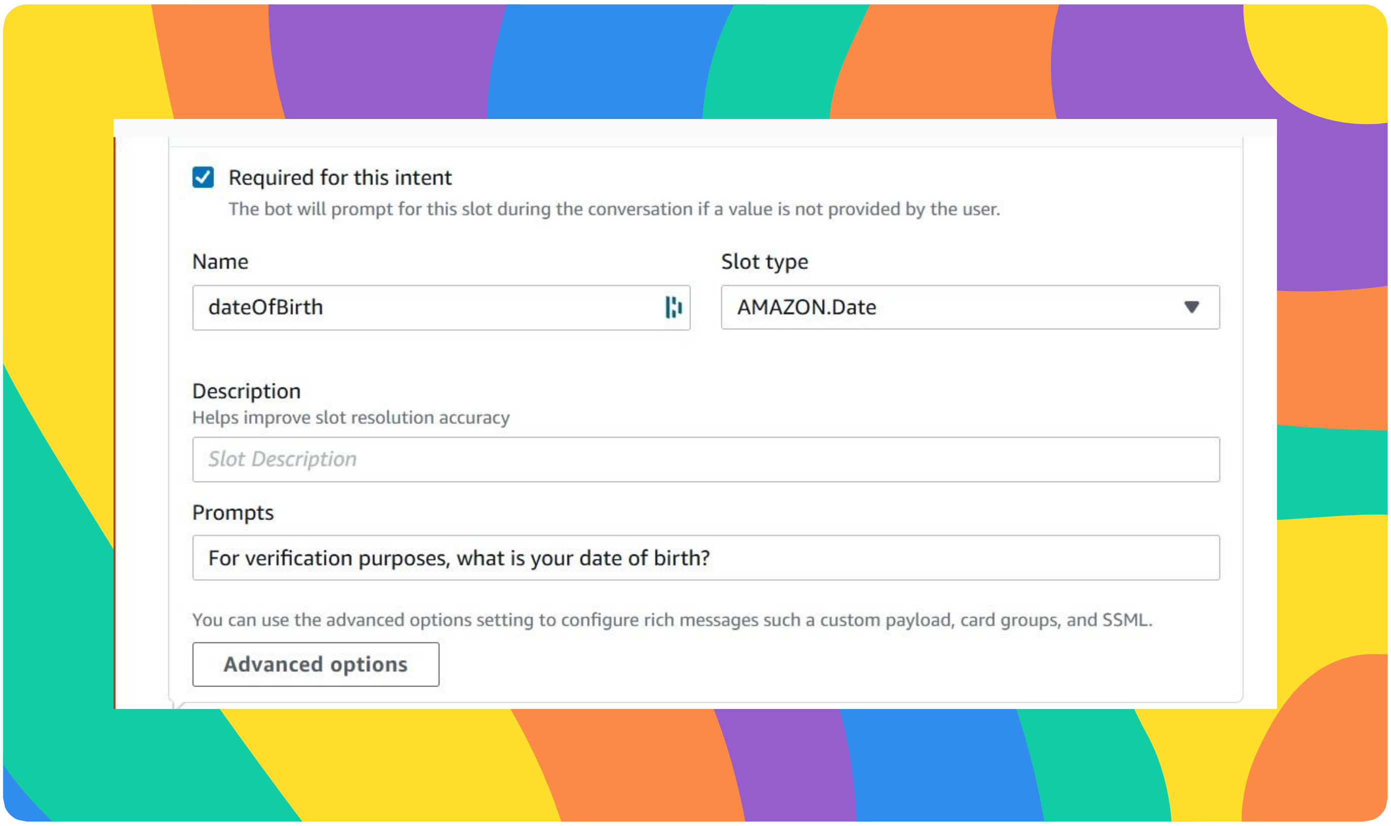
This project took me a few focused hours, during which I designed the intents, created custom slot types, configured prompts and failure responses, and tested the full conversational flow to ensure the chatbot behaved naturally and reliably.

Slots

Slots are the pieces of information a chatbot collects during a conversation. They work like blanks in a form that the user fills in so the bot can complete their request.

By adding my custom slots to the intent's sample utterances, users can provide inputs like savings, checking, or credit. These responses match the custom slot values I defined, allowing Lex to capture the correct account type during the conversation.

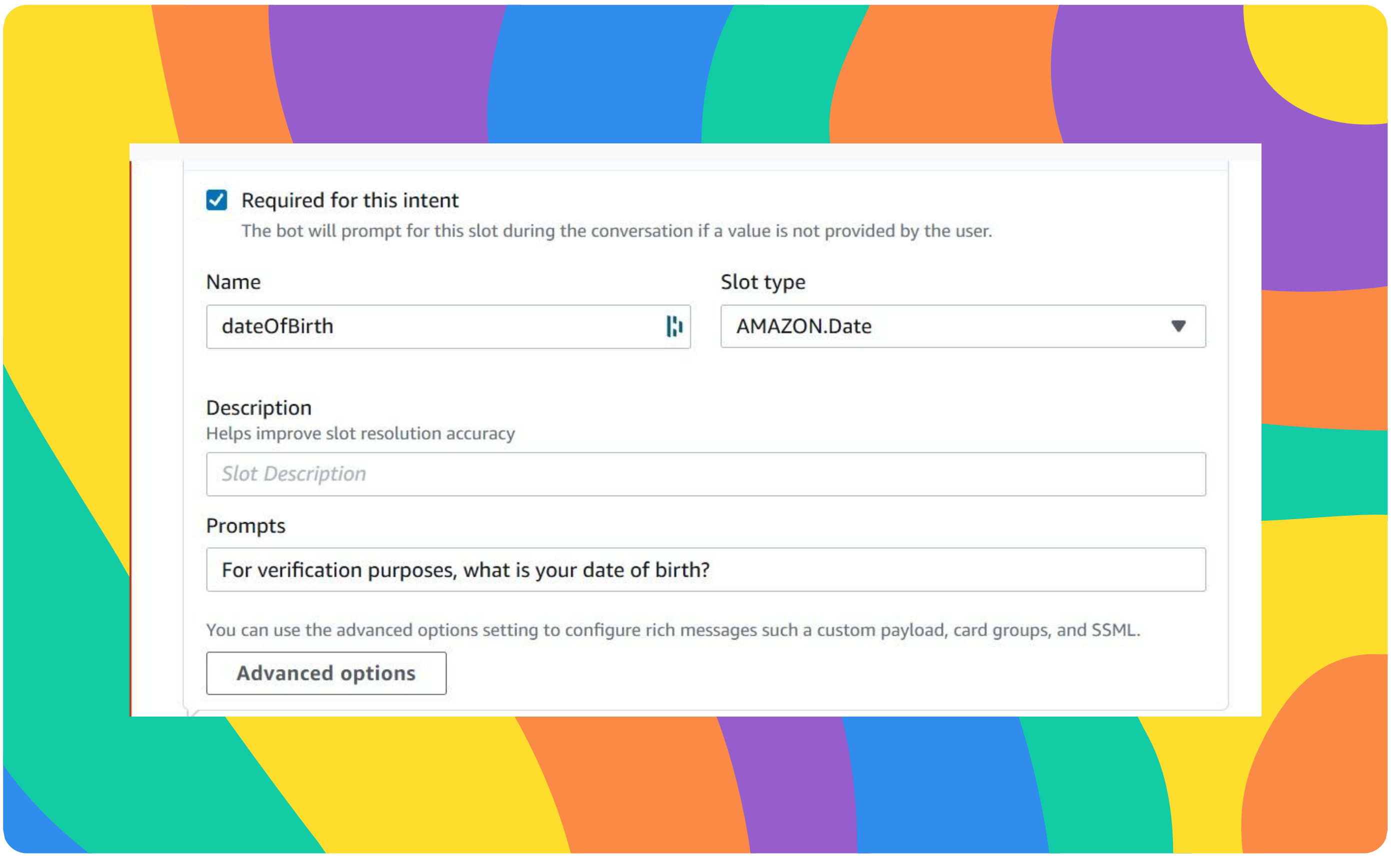
For this project, I created a custom slot type to ensure Lex only accepts the account types supported by my imaginary bank. By defining values like checking, savings, and others, Lex can reliably capture the user's account type and use it to process their banking request correctly.



Connecting slots with intents

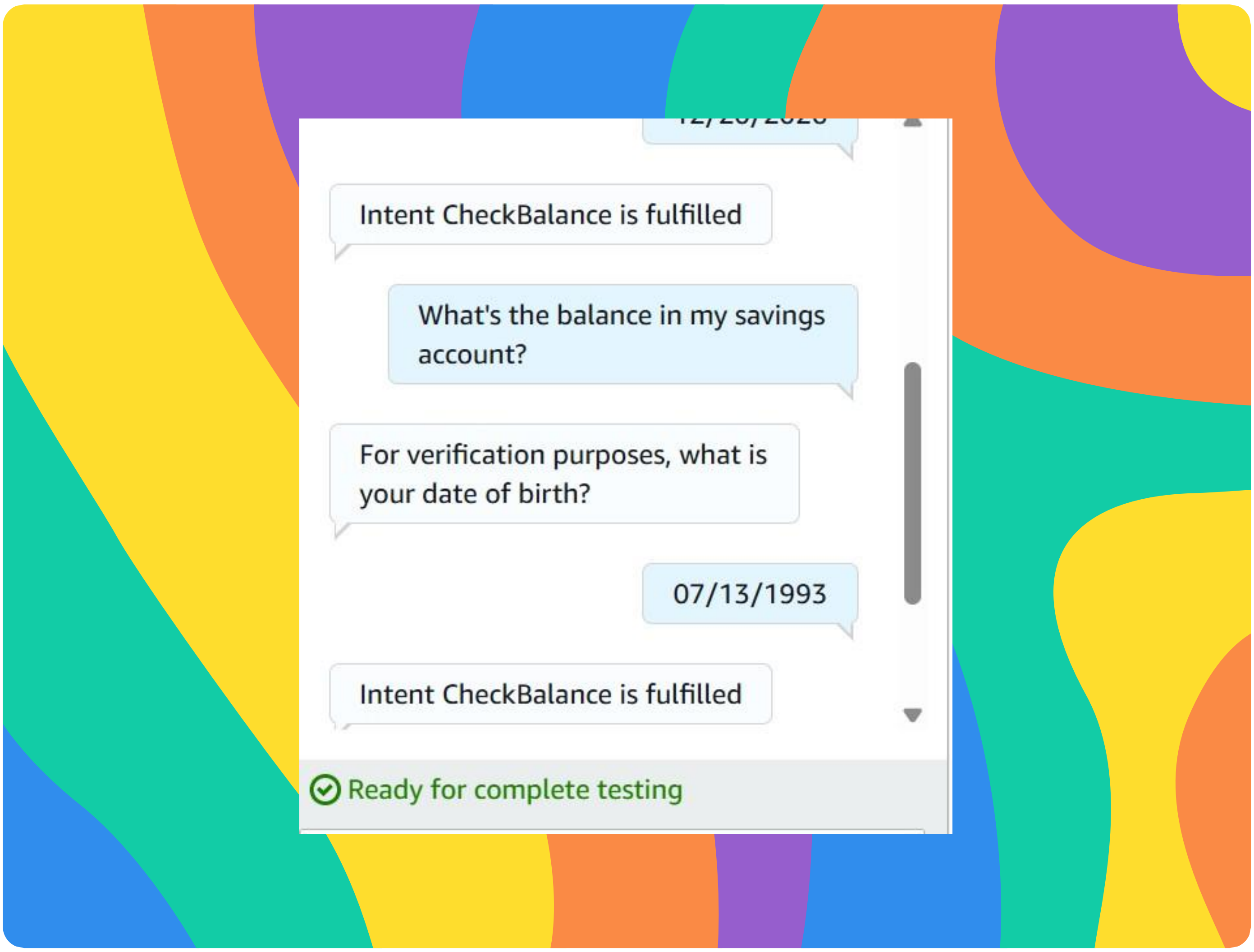
This slot type uses restricted slot values, which means Amazon Lex will only accept the exact values I define and reject anything else. Without this restriction, Lex might rely on machine learning to accept other values it frequently sees from users, even if they aren't valid account types for my chatbot.

I associated my custom slot with the CheckBalance intent, which is responsible for collecting the user's information so the bot can check their bank balance. By linking the slot to this intent, Lex knows to capture the user's account type as part of the conversation flow..



Slot values in utterances

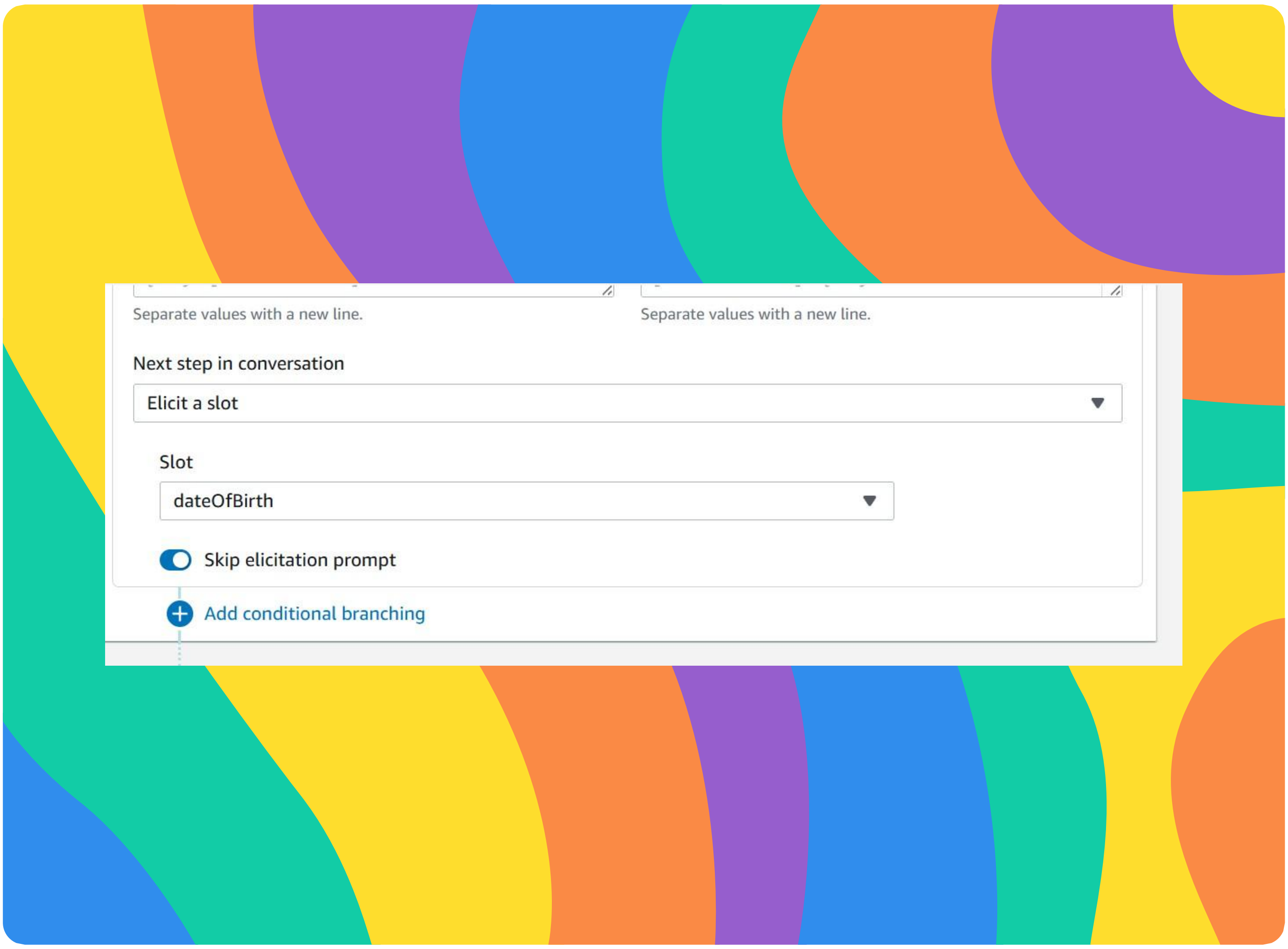
I included my custom slot in some of the sample utterances by using curly braces. For example: “What’s the balance in my {accountType} account?” This allows Lex to capture the user’s account type directly from their phrasing during the conversation.



Handling failures in slot values

I added variations for the dateOfBirth slot prompt, such as “Umm, that didn’t work either. Try sharing your date of birth in MM/DD/YY.” and “Sorry, that wasn’t clear to me. What’s your date of birth?” These variations help the bot sound more natural and guide the user when their input doesn’t match the expected format.

I used a failure response to handle invalid user inputs and changed the default settings to clarify that we are expecting a birthdate for verification.


Separate values with a new line.
Separate values with a new line.
Next step in conversation
Elicit a slot
Slot
dateOfBirth
Skip elicitation prompt
Add conditional branching